A) Building a DAO from Scratch – The Customer Dao

Note: This uses the classicmodels database (from our previous sample code).

Part 1) Create a Record to model the customers table

Use Java's record keyword to define a new Customer record type. Remember to override equals() and hashCode() to only include the primary key for the customers table.

Part 2) Create the Customer DAO interface

i. Write a java interface called `CustomerDao`. This interface should specify the following methods:
   a. `List<Customer> selectCustomersByName(String name);`
      Get all Customers matching the supplied name exactly
   b. `List<Customer> selectCustomersContainingName(String name);`
      Get all Customers with a customer name **containing** the supplied text
   c. `Customer findCustomerById(int customerNumber);`
      Get the Customer matching the specified customer id number
   d. `boolean addCustomer(Customer c);`
      Add the supplied Customer object to the database (return true if successful & false otherwise).

Part 3) Create the CustomerDaoImpl class – providing access to the Customer table within the database

i. Write a class called *CustomerDaoImpl* that implements *CustomerDao*. The class should implement the *CustomerDao* fully, i.e. provide implementations for the methods listed in the interface.

Part 4) Write applications using DAO functionality

i. Create a java program that takes in a piece of text from the user and displays all Customers **matching** that name. Where there are no matches, your program should check for any Customers ***containing*** the text supplied and display those.

ii. Create a separate java program that **repeatedly** allows a user to check if a customer id number is available. Once they have found an available id number, prompt the user to enter the remaining required information for the new Customer and add it to the database.